

LINEAR ALGEBRA BASICS FOR MACHINE LEARNING

Linear Algebra is one of the **most important mathematical foundations** of Machine Learning, Deep Learning, Computer Vision, and Artificial Intelligence.

Almost every ML algorithm uses Linear Algebra internally.

WHAT IS LINEAR ALGEBRA?

Linear Algebra is the branch of mathematics that deals with

- Numbers
- Vectors
- Matrices
- Linear Equations

It helps computers store and manipulate data efficiently.

Simple Definition

Linear Algebra is the mathematics of vectors and matrices.

Think of it as the language computers use to perform machine learning calculations.

WHY DO WE NEED LINEAR ALGEBRA?

Suppose you have a dataset.

Student Age Study Hours Marks

A	18	2	40
B	19	4	55
C	20	6	70

To a computer, this is represented as a **matrix**:

```
18 2 40
19 4 55
20 6 70
```

Machine learning algorithms work with this numeric matrix, not with tables as humans see them.

WHERE IS LINEAR ALGEBRA USED IN MACHINE LEARNING?

ML Algorithm	Linear Algebra Used?
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Linear Regression	✓ Yes
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Logistic Regression	✓ Yes
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Neural Networks	✓ Yes
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PCA	✓ Yes
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SVM	✓ Yes
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K-Means	✓ Yes
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Deep Learning	✓ Yes
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Computer Vision	✓ Yes
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MAIN TOPICS OF LINEAR ALGEBRA

1. Scalars
2. Vectors
3. Matrices
4. Matrix Operations
5. Transpose
6. Dot Product
7. Matrix Multiplication
8. Identity Matrix
9. Inverse Matrix
10. Determinant
11. Eigenvalues & Eigenvectors (Advanced)
12. Singular Value Decomposition (Advanced)

Let's learn the basics first.

1. SCALAR

A **scalar** is just a single number.

Examples

5
10
-3
2.5
100

Python

```
x = 5  
print(x)
```

Output

5

ML Example

Learning Rate

0.01

Epochs

100

These are scalars.

2. VECTOR

A vector is a collection of numbers arranged in one dimension.

Example

[2,4,6,8]

This could represent

Study Hours of 4 Students

Python

```
import numpy as np
```

```
v = np.array([2,4,6,8])
```

```
print(v)
```

Output

```
[2 4 6 8]
```

WHY ARE VECTORS USED?

Suppose

Student Marks

80

75

90

60

Instead of storing them separately,

we store

```
[80,75,90,60]
```

This is much faster.

3. MATRIX

A matrix is a collection of rows and columns.

Example

```
18  2  40
```

```
19  4  55
```

```
20  6  70
```

Python

```
import numpy as np
```

```
A = np.array([[18,2,40],  
              [19,4,55],  
              [20,6,70]])
```

```
print(A)
```

Output

```
[[18  2 40]
```

```
 [19  4 55]
```

```
 [20  6 70]]
```

WHY ARE MATRICES USED?

Every dataset is converted into a matrix before training.

Example

Age Hours Marks

18 2 40

19 4 55

20 6 70

becomes

18 2 40

19 4 55

20 6 70

Matrix Shape

Shape means

Rows × Columns

Example

1 2 3

4 5 6

Rows = 2

Columns = 3

Shape

2 × 3

Python

```
import numpy as np
```

```
A = np.array([[1,2,3],  
              [4,5,6]])
```

```
print(A.shape)
```

Output

(2,3)

MATRIX ADDITION

Both matrices must have the same size.

Example

A

1 2
3 4

B

5 6
7 8

Addition

6 8
10 12

Python

```
import numpy as np
```

```
A = np.array([[1,2],[3,4]])
```

```
B = np.array([[5,6],[7,8]])
```

```
print(A+B)
```

MATRIX SUBTRACTION

```
print(A-B)
```

Output

[-4 -4]
[-4 -4]

MATRIX MULTIPLICATION

Matrix multiplication is one of the most important operations in Machine Learning.

Suppose

A

1 2
3 4

B

5 6
7 8

Each element of the result is found by multiplying a row of **A** with a column of **B**.

×

$$\begin{bmatrix} 2 & 1 & 3 \\ 4 & 0 & 2 \end{bmatrix} \times \begin{bmatrix} 5 & 2 \\ 1 & 4 \\ 3 & 7 \end{bmatrix} = \begin{bmatrix} 20 & 29 \\ 26 & 22 \end{bmatrix}$$

$(AB)_{11} = (2 \cdot 5) + (1 \cdot 1) + (3 \cdot 3) = 20$

Row	1	2
Column	1	2

Python

```
import numpy as np
```

```
A = np.array([[1,2],  
              [3,4]])
```

```
B = np.array([[5,6],  
              [7,8]])
```

```
print(np.matmul(A,B))
```

Output

```
[[19 22]  
 [43 50]]
```

WHY MATRIX MULTIPLICATION IS USED IN ML

Neural Networks perform millions of matrix multiplications every second.

Example

Input Matrix

×

Weight Matrix

=

Prediction

Every AI model uses this operation.

TRANSPOSE OF A MATRIX

Rows become columns.

Example

A

1 2 3
4 5 6

Transpose

1 4
2 5
3 6

Python

```
import numpy as np
```

```
A = np.array([[1,2,3],  
              [4,5,6]])
```

```
print(A.T)
```

Output

```
[[1 4]  
 [2 5]  
 [3 6]]
```

WHY TRANSPOSE IS USED

Used in

- Linear Regression

- PCA
 - Neural Networks
 - Covariance Matrix
-

DOT PRODUCT

The dot product measures how similar two vectors are.

Example

$A = [2, 3]$

$B = [4, 5]$

$2 \times 4 + 3 \times 5$

$= 8 + 15$

$= 23$

Python

```
import numpy as np
```

```
A=np.array([2,3])
```

```
B=np.array([4,5])
```

```
print(np.dot(A,B))
```

Output

23

IDENTITY MATRIX

Diagonal elements are 1.

Others are 0.

Example

1 0 0

0 1 0

0 0 1

Python

```
import numpy as np
```

```
print(np.eye(3))
```

Output

```
[[1. 0. 0.]  
 [0. 1. 0.]  
 [0. 0. 1.]]
```

WHY IDENTITY MATRIX?

Similar to multiplying a number by 1.

$$5 \times 1 = 5$$

Similarly,

$$A \times I = A$$

INVERSE MATRIX

Inverse is similar to division.

Example

$$A \times A^{-1} = I$$

Python

```
import numpy as np
```

```
A=np.array([[1,2],  
            [3,4]])
```

```
print(np.linalg.inv(A))
```

Determinant

Determines whether a matrix is invertible.

Python

```
import numpy as np
```

```
A=np.array([[1,2],  
            [3,4]])
```

```
print(np.linalg.det(A))
```

Output

-2

If the determinant is **0**, the matrix has **no inverse**.

HOW LINEAR ALGEBRA IS USED IN ML

1. Linear Regression

Finds the best-fit line using matrix operations.

2. NEURAL NETWORKS

- Inputs → Vectors
- Weights → Matrices
- Predictions → Matrix Multiplication

3. IMAGE PROCESSING

A grayscale image is stored as a matrix.

Example

```
120 130 140
110 125 150
100 115 145
```

Each value is the intensity of a pixel.

4. NATURAL LANGUAGE PROCESSING (NLP)

Each word is converted into a numeric vector.

Example

"cat"

↓

[0.25, 0.91, 0.42, ...]

These are called **word embeddings**.

5. PRINCIPAL COMPONENT ANALYSIS (PCA)

PCA uses matrices and eigenvectors to reduce the number of features while preserving most of the important information.

NUMPY FUNCTIONS USED IN LINEAR ALGEBRA

Function	Purpose
<code>np.array()</code>	Create vectors/matrices
<code>shape</code>	Matrix dimensions
<code>A.T</code>	Transpose
<code>np.dot()</code>	Dot product
<code>np.matmul()</code> or <code>@</code>	Matrix multiplication
<code>np.eye()</code>	Identity matrix
<code>np.linalg.inv()</code>	Inverse matrix
<code>np.linalg.det()</code>	Determinant
<code>np.linalg.eig()</code>	Eigenvalues and eigenvectors (advanced)

Summary

Concept	Meaning	ML Use
Scalar	Single number	Learning rate, bias
Vector	1D list of numbers	Features, weights
Matrix	2D table of numbers	Datasets, images
Transpose	Swap rows and columns	Regression, PCA
Dot Product	Multiply and sum vectors	Similarity, predictions
Matrix Multiplication	Multiply matrices	Neural networks, regression
Identity Matrix	Matrix version of 1	Matrix algebra

Concept	Meaning	ML Use
Inverse Matrix	Matrix version of division	Solving equations
Determinant	Checks invertibility	Matrix analysis